

# Combining Technical Indicators and Equal-Budget Metaheuristic Optimization for Short-Term Mini-Index Futures Prediction

## Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound  $0.60 \text{ MCC} + 0.40 F_1$  fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

**Keywords:** Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

## 1 Introduction

Financial markets provide a setting in which computational prediction must contend with noisy observations, short-lived regularities, and changing market regimes. Futures contracts are particularly relevant because they combine high liquidity with rapid reactions to macroeconomic news, order flow, and changes in perceived risk. The Brazilian Mini-Index futures contract (WIN), referenced to the Ibovespa index, is consequently an appropriate instrument for studying short-horizon intraday trend classification.

Five-minute price series contain information about local momentum, dispersion, and trading activity, but they also contain substantial microstructure noise. Technical indicators, including moving averages, exponential moving averages, and volatility transformations, remain a practical way to convert raw prices into supervised-learning descriptors [1, 2]. Their usefulness depends on avoiding redundant inputs and on evaluating candidate models without allowing future observations to influence the selection process.

Machine-learning methods provide flexible nonlinear decision functions for this problem. Random Forest (RF) and Support Vector Machine (SVM) classifiers are established baselines, whereas Multilayer Perceptrons (MLPs) and one-dimensional convolutional neural networks (1D-CNNs) can represent nonlinear interactions between technical descriptors. In all four cases, predictive behaviour depends materially on architecture and training hyperparameters. Metaheuristics are therefore useful because the search space combines continuous, integer, and categorical decisions that are difficult to set manually [3–6].

The conference precursor to this work combined technical indicators, Information Gain feature selection, and genetic-algorithm (GA) tuning for RF, SVM, and MLP models [7]. Its central idea remains useful: a compact technical representation can make evolutionary model selection feasible for intraday data. The present study changes the experimental question. Rather than considering only GA-tuned models under randomized cross-validation, it compares Random Search (RS), GA, Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimization (GWO) under a common chronological protocol and an identical evaluation allowance.

The objective is to determine how the choice of validation objective interacts with model and optimizer selection on a locked, chronological test segment. Two controlled protocols are considered: an imbalance-aware compound  $MCC/F_1$  fitness and a weighted training/validation accuracy fitness. The test block is kept inaccessible during feature selection, candidate ranking, and hyperparameter search. This separation is essential because high validation performance does not, by itself, establish robust out-of-sample predictive or economic utility [8, 9].

This manuscript makes six contributions:

- **Preserves** the precursor’s Information-Gain-selected seven-indicator representation for WIN futures.
- **Replaces** randomized validation with a chronological 60/20/20 train/validation/locked-test protocol.
- **Compares** RS, GA, PSO, DE, and GWO across RF, SVM, MLP, and 1D-CNN backbones.
- **Controls** optimizer opportunity through 1,000 candidate evaluations per optimizer and seed.
- **Separates**  $MCC/F_1$  and weighted accuracy fitness functions from the final reported test metrics.
- **Extends** predictive evaluation with exploratory Friedman–Wilcoxon analysis and an intraday economic backtest.

The remainder of the article follows the organizational pattern of the precursor. Section 2 reviews related work. Section 3 describes the data, technical indicators, feature selection, and optimization design. Section 4 presents the experimental setup, model results, statistical analysis, and economic evaluation. Section 5 concludes the study.

## 2 Related Work

The use of machine learning and deep learning for financial prediction has expanded as public and proprietary market datasets have become more accessible. Financial series remain challenging because apparent regularities may be temporary and because predictive evaluation can be distorted by an inappropriate temporal split [10, 11]. Studies comparing learning algorithms therefore commonly report that model choice, feature representation, and validation design jointly affect the observed ranking.

Technical analysis indicators are a common bridge between raw market prices and supervised models. Moving-average and volatility descriptors summarize short-run trend and dispersion, but a large indicator set may introduce redundancy and unstable fitted models. Information-theoretic filtering is a simple and reproducible route to retain attributes that reduce uncertainty about the target class [12]. The precursor adopted this strategy to reduce 66 engineered descriptors to seven inputs before fitting RF, SVM, and MLP classifiers [7].

Hyperparameter optimization is equally important when neural, kernel, and ensemble models are applied to financial data. GA provides a population-based global search mechanism; PSO exploits individual and collective attraction; DE uses differential perturbations; and GWO uses a hierarchy of leading agents [3–6]. RS is also necessary in a controlled comparison because it supplies a transparent budget-matched baseline [13]. These methods cannot be compared fairly when one receives more candidate evaluations than another.

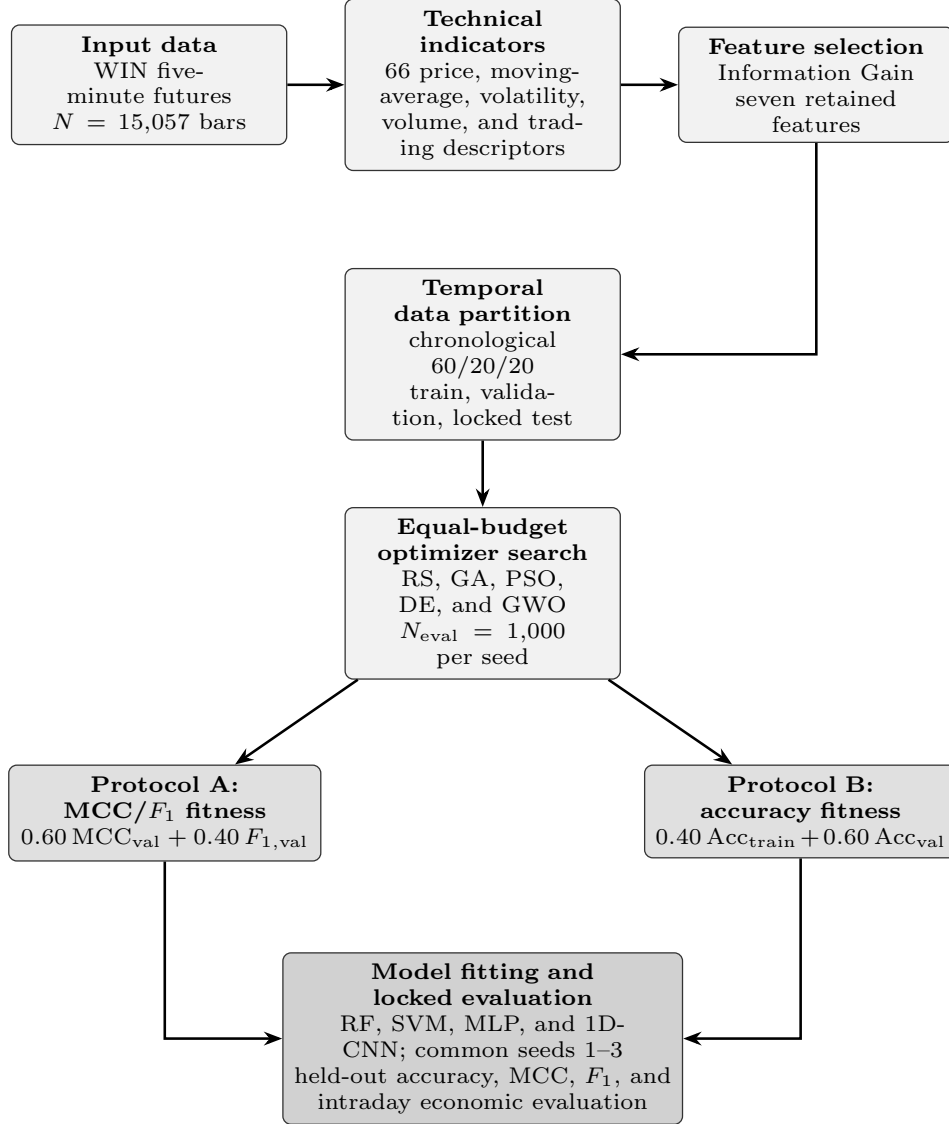
Recent financial-learning studies have also emphasized temporal evaluation and economic relevance. A validation score selected on one period need not transfer to a later regime, and classification accuracy alone cannot specify the return-risk trade-off of a trading rule. MCC is useful when class balance and both types of classification error matter, while trading evaluation additionally requires profit and drawdown measures [8, 14].

Unlike the precursor, the present work integrates feature selection, equal-budget multi-optimizer search, two validation fitness functions, locked chronological testing, and economic evaluation in one protocol. The extension is deliberately conservative: direct comparisons between the two fitness functions are descriptive because only three common seeds are available in every model–optimizer cell.

## 3 Methodology

This section presents the methodology adopted in this study. Figure 1 provides an overview of the flow from data acquisition to final model evaluation. It preserves the practical sequence of the IJCNN precursor—input data, technical indicators, feature

selection, optimization, model fitting, and evaluation—while inserting the temporal controls and the two validation objectives required by the expanded benchmark.

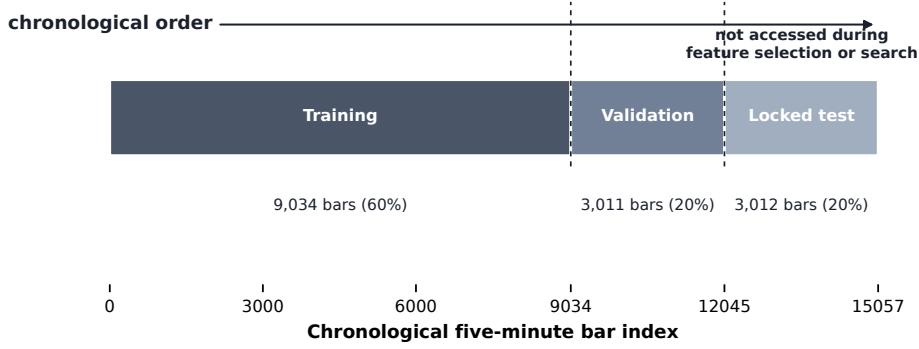


**Fig. 1** Overview of the proposed methodology. The new benchmark preserves the precursor workflow while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

### 3.1 Input Data

The data were obtained from the BovDB resources [15, 16]. The analyzed subset contains 15,057 clean five-minute records of the WIN contract from January to June 2024. WIN is referenced to the Ibovespa index and was selected because its liquidity and frequent intraday movement make it suitable for short-horizon computational prediction. The binary target indicates the next-bar direction: 0 for downtrend and 1 for uptrend.

The records remain in chronological order. The first 9,034 observations are used for training, the subsequent 3,011 observations for validation, and the final 3,012 observations for locked testing. No random shuffling is applied. Figure 2 makes this separation explicit; the test observations are unavailable during both feature and hyperparameter selection.



**Fig. 2** Chronological data partition used in every experiment. Training, validation, and locked test segments occupy consecutive periods of the WIN five-minute series.

### 3.2 Generation of Technical Analysis Indicators

Technical indicators were derived from open, high, low, close, trade volume, and transaction-frequency information. Simple Moving Average (SMA), Exponential Moving Average (EMA), and dispersion measures were computed over short windows (3, 5, 7, and 9 periods) to retain sensitivity to intraday changes. For example, the difference between the seven- and three-period SMA is

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3. \quad (1)$$

The raw and derived variables were normalized before model fitting. Together, the construction procedure produces 66 descriptors, including moving-average differences and normalized counterparts.

### 3.3 Feature Selection Method

The indicator pool is reduced using Information Gain, as in the precursor. This filter selects attributes that most reduce uncertainty regarding the next-bar direction and avoids repeating highly redundant descriptors in each search evaluation [12, 17]. Seven features are retained:  $\text{EMA}_{5-3}$ , normalized  $\text{EMA}_{5-3}$ ,  $\text{EMA}_{7-3}$ , normalized  $\text{EMA}_{7-3}$ , normalized  $\text{SMA}_9$ ,  $\text{EMA}_{9-3}$ , and normalized  $\text{EMA}_7$ . The same reduced representation is used by all model and optimizer combinations.

### 3.4 Optimizer Configuration and Validation Objectives

Five optimizers are evaluated: RS, GA, PSO, DE, and GWO. Each optimizer receives 1,000 candidate evaluations for each model and seed. Population-based methods use populations of ten candidates, corresponding to 100 generations or iterations; RS samples 1,000 feasible configurations. GA uses tournament selection, crossover, and mutation; PSO uses inertia plus cognitive and social terms; DE uses the best/1/bin strategy; and GWO updates candidate positions from its leading agents.

Two validation objectives are evaluated. Protocol A uses the compound metric

$$f_{\text{MCC}/F1}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}), \quad (2)$$

whereas Protocol B uses

$$f_{\text{Acc}}(\boldsymbol{\theta}) = 0.40 \text{Accuracy}_{\text{train}}(\boldsymbol{\theta}) + 0.60 \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (3)$$

After candidate selection, the selected configuration is refit on training plus validation data and evaluated once on the locked test block.

## 4 Results

This section presents the experimental results and their implications for short-term prediction of Brazilian Mini-Index futures. Following the organization of the precursor, it first describes the common experimental setup and then discusses the performance of the individual model families. The expanded benchmark subsequently reports the direct cross-model comparison, exploratory statistical evidence, and economic outcomes.

### 4.1 Experimental Setup and Data Partitioning

All model–optimizer combinations use the seven descriptors described in Section 3.3, the chronological split shown in Figure 2, and the same 1,000-evaluation allowance per seed. RF, SVM, MLP, and 1D-CNN each are searched with RS, GA, PSO, DE, and GWO. The two protocols differ only in the validation fitness used to rank candidate configurations. Three common seeds (1–3) are available for every cell in both protocols.

Consequently, the direct comparison between fitness functions is descriptive: each reported model–optimizer result is the mean over the same three seeds. The locked test block contains 3,012 observations and is not used during search. This design differs

from the randomized training and test sampling used in the precursor and removes look-ahead exposure from model selection.

## 4.2 Performance of the Models

Tables 1–4 retain the presentation pattern of the precursor: each table states the search space and an illustrative GA-selected configuration. The configurations are from seed 1 under Protocol A and are included to make the optimization procedure concrete; they are not asserted to be universally optimal because all optimizer–seed outcomes are summarized below.

**Table 1** Random Forest hyperparameters and representative GA-selected configuration.

| Hyperparameter          | Search space       | Selected value |
|-------------------------|--------------------|----------------|
| $n_{\text{estimators}}$ | [50, 500]          | 58             |
| max_depth               | [5, 50]            | 6              |
| min_samples_split       | [2, 20]            | 13             |
| min_samples_leaf        | [1, 20]            | 15             |
| max_features            | {sqrt, log2, None} | None           |
| class_weight            | {None, balanced}   | None           |

**Table 2** SVM hyperparameters and representative GA-selected configuration.

| Hyperparameter | Search space                  | Selected value |
|----------------|-------------------------------|----------------|
| $C$            | $[10^{-3}, 10^3]$ (log scale) | 2.089          |
| $\gamma$       | $[10^{-5}, 10^1]$ (log scale) | 5.044          |
| Kernel         | {linear, RBF}                 | linear         |

**Table 3** MLP hyperparameters and representative GA-selected configuration.

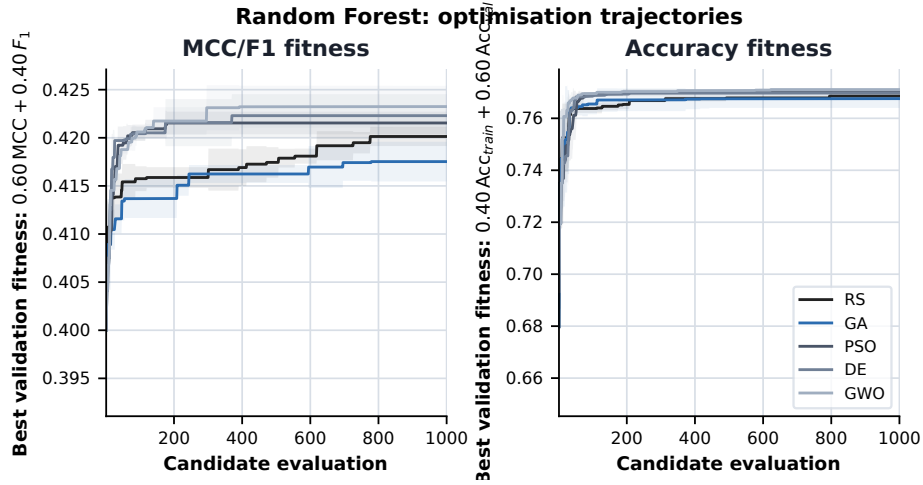
| Hyperparameter       | Search space                     | Selected value        |
|----------------------|----------------------------------|-----------------------|
| Hidden neurons       | [5, 500]                         | 410                   |
| Learning rate        | $[10^{-8}, 10^{-2}]$ (log scale) | $3.42 \times 10^{-4}$ |
| $L_2$ regularization | $[10^{-8}, 10^{-2}]$ (log scale) | $1.66 \times 10^{-7}$ |
| Dropout rate         | [0, 0.10]                        | 0.057                 |
| Batch size           | {16, 32, 64, 80, 112, 128}       | 112                   |

Figures 3–6 replace the three GA-only convergence plots of the precursor with the five budget-matched search trajectories. For each model, the left panel follows the

**Table 4** 1D-CNN hyperparameters and representative GA-selected configuration.

| Hyperparameter       | Search space                     | Selected value        |
|----------------------|----------------------------------|-----------------------|
| Filters              | [8, 128]                         | 48                    |
| Kernel size          | [2, 5]                           | 3                     |
| Dense neurons        | [16, 256]                        | 251                   |
| Learning rate        | $[10^{-8}, 10^{-2}]$ (log scale) | $3.90 \times 10^{-6}$ |
| $L_2$ regularization | $[10^{-8}, 10^{-2}]$ (log scale) | $5.70 \times 10^{-6}$ |
| Dropout rate         | [0, 0.10]                        | 0.026                 |
| Batch size           | {16, 32, 64, 80, 112, 128}       | 16                    |

compound  $MCC/F_1$  fitness and the right panel follows weighted training/validation accuracy. The vertical scales differ because the objectives differ; their role is to show search convergence within a protocol, not to compare objective magnitudes across protocols.



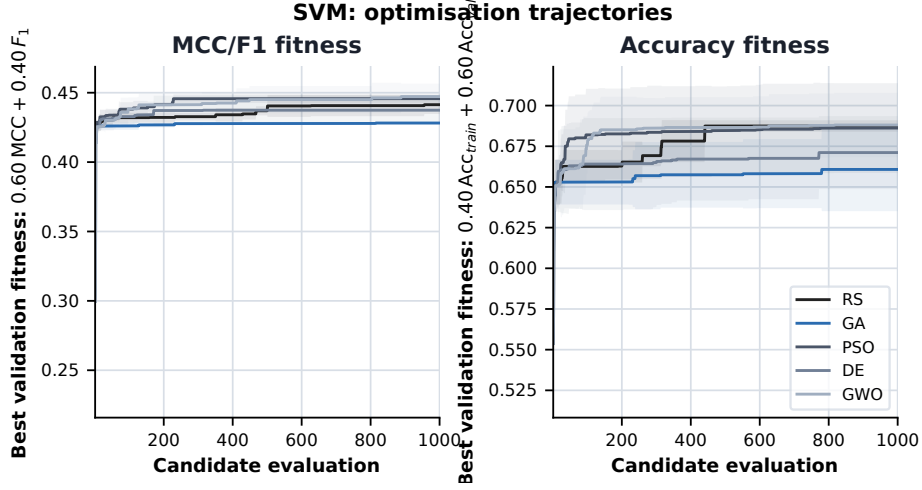
**Fig. 3** Fitness evolution during equal-budget optimization of the Random Forest model. Lines summarize the three common seeds.

Table 5 and Figure 7 present held-out performance aggregated over 15 optimizer-seed blocks per model within each protocol. Under  $MCC/F_1$  fitness, RF has the largest mean test accuracy (0.645),  $MCC$  (0.289), and  $F_1$  (0.664). Under weighted accuracy fitness, MLP has the largest mean test accuracy (0.651),  $MCC$  (0.301), and  $F_1$  (0.658). The reversal reinforces that the validation criterion and model family should be reported separately rather than treated as interchangeable.

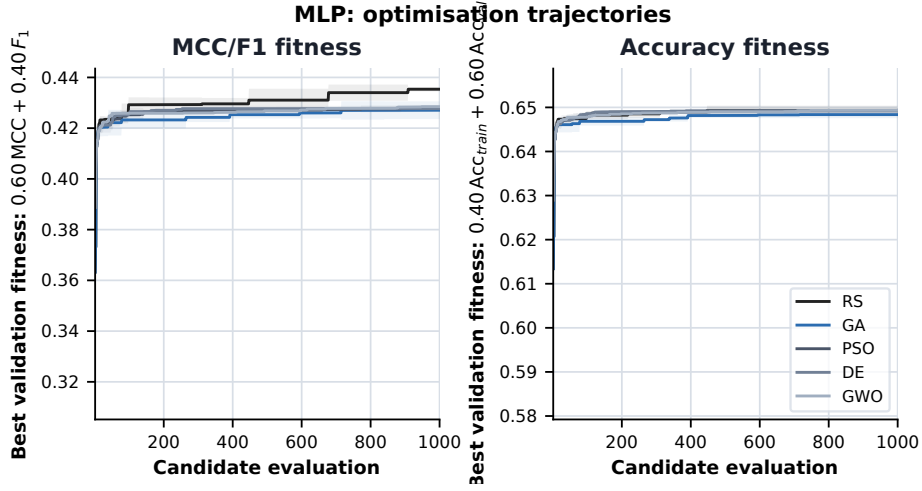
### 4.3 Exploratory Statistical and Economic Evaluation

The model-level tests are performed within each holdout protocol. The five optimizer-seed combinations yield 15 matched blocks for each backbone. Under  $MCC/F_1$  fitness,





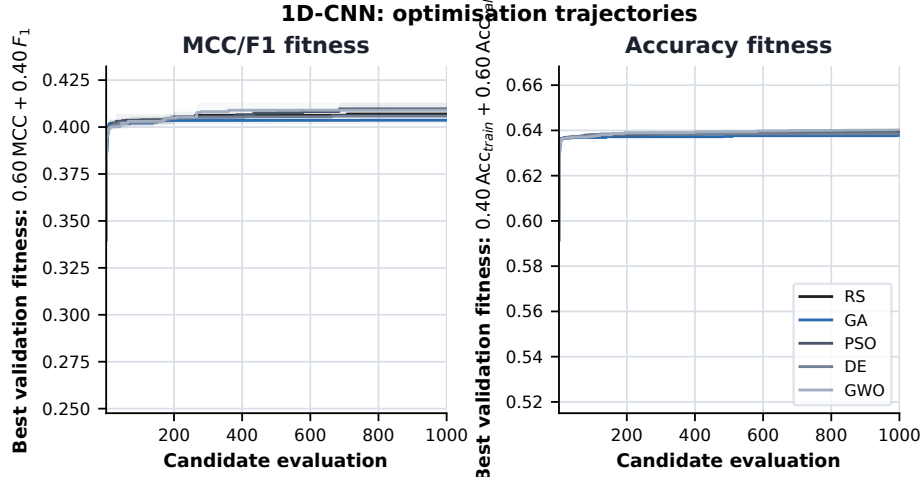
**Fig. 4** Fitness evolution during equal-budget optimization of the SVM model. Lines summarize the three common seeds.



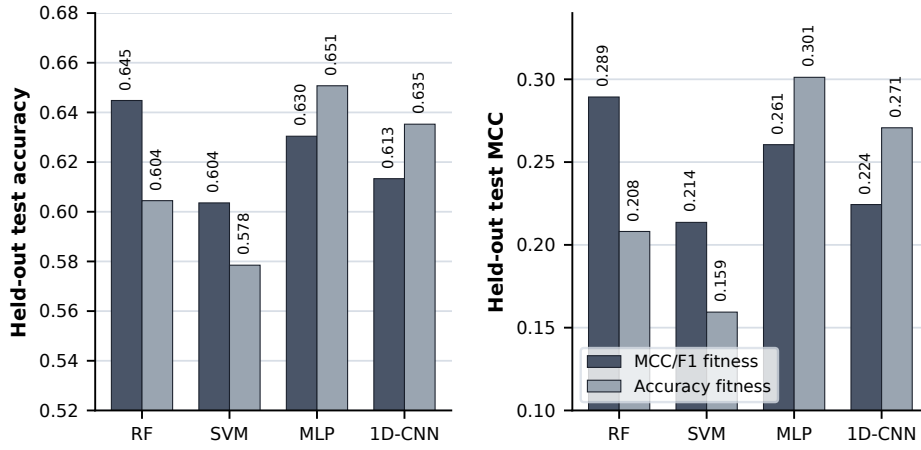
**Fig. 5** Fitness evolution during equal-budget optimization of the MLP model. Lines summarize the three common seeds.

**Table 5** Mean locked-test predictive outcomes by backbone and validation protocol (15 optimizer-seed blocks per row).

| Model  | MCC/ $F_1$ fitness |       |       | Weighted accuracy fitness |       |       |
|--------|--------------------|-------|-------|---------------------------|-------|-------|
|        | Accuracy           | MCC   | $F_1$ | Accuracy                  | MCC   | $F_1$ |
| RF     | 0.645              | 0.289 | 0.664 | 0.604                     | 0.208 | 0.624 |
| SVM    | 0.604              | 0.214 | 0.602 | 0.578                     | 0.159 | 0.600 |
| MLP    | 0.630              | 0.261 | 0.639 | 0.651                     | 0.301 | 0.658 |
| 1D-CNN | 0.613              | 0.224 | 0.597 | 0.635                     | 0.271 | 0.639 |



**Fig. 6** Fitness evolution during equal-budget optimization of the 1D-CNN model. Lines summarize the three common seeds.

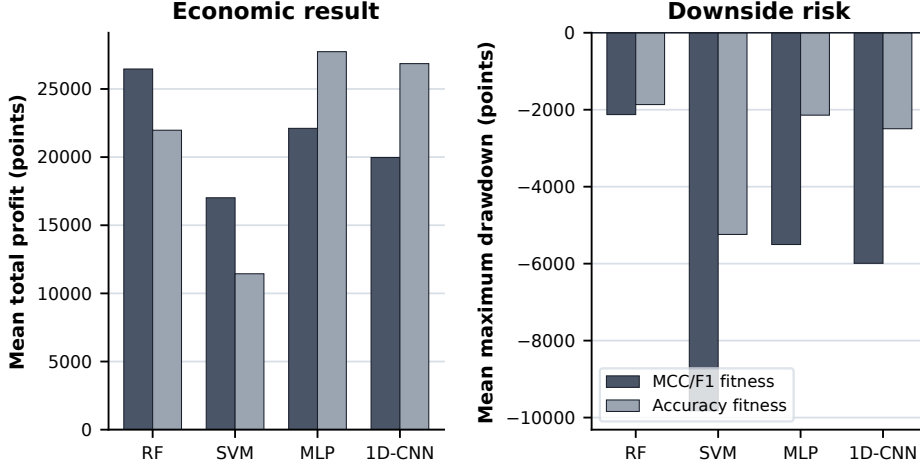


**Fig. 7** Comparison of mean locked-test accuracy and MCC for all four model families under the two validation objectives.

the Friedman comparison of test MCC is significant ( $\chi^2 = 22.84$ ,  $p = 4.36 \times 10^{-5}$ ), with RF obtaining the best mean outcome. Under weighted accuracy fitness, the comparison of test accuracy is significant ( $\chi^2 = 42.92$ ,  $p = 2.56 \times 10^{-9}$ ), with MLP first; its Holm-adjusted pairwise contrasts against RF, SVM, and 1D-CNN are significant ( $p_{\text{Holm}} \approx 0.00194$ ). These tests compare models within a protocol and should not be interpreted as a confirmatory test between the two fitness functions.

The economic evaluation yields a complementary ranking. Under MCC/ $F_1$  fitness, RF has the largest mean total profit (26,465 points). Under weighted accuracy fitness, MLP has the largest mean total profit (27,737 points), whereas RF has the least

negative mean maximum drawdown (-1,866 points). Figure 8 therefore makes explicit the difference between predictive ranking and return-risk ranking.



**Fig. 8** Mean total profit and maximum drawdown across 15 optimizer-seed blocks per model. Results are reported separately for the two validation objectives.

## 5 Conclusion

This work extends the IJCNN precursor on technical indicators, Information Gain feature selection, and GA-based hyperparameter optimization for Mini-Index futures prediction. The original practical pipeline is retained: a compact set of technical descriptors is constructed from intraday WIN data, informative attributes are selected, and nonlinear models are optimized before final testing. The present study expands that design through a chronological train/validation/test split, four model families, five budget-matched optimizers, two explicit validation objectives, and a locked economic evaluation.

The results show that a single validation score cannot define a universal model ranking. RF produces the strongest mean locked-test accuracy, MCC, and  $F_1$  under the compound MCC/ $F_1$  fitness, whereas MLP leads the predictive results under weighted accuracy fitness. The economic analysis reaches a related but distinct conclusion: MLP has the largest mean profit under weighted accuracy fitness, while RF has the smallest mean drawdown. These findings support the use of controlled metaheuristic optimization, but they also show why the selection objective, reported predictive metric, and economic endpoint must be stated separately.

The direct comparison between fitness functions remains descriptive because only three seeds are common to every model-optimizer cell. The statistical analysis is correspondingly limited to exploratory within-protocol comparisons. Future work should complete a preregistered common 20-seed design, retain the same temporal split and evaluation budget, and repeat the analysis across additional liquid futures markets

and market regimes. Such an extension would test the stability of the present results while preserving the transparent workflow established here.

## Declarations

### Funding

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### Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

### Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

### Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

### Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

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